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Comparison of the removal efficiency for organic contaminants on silicon wafers stored in plastic boxes between UV/O_3 and ECR oxygen plasma cleaning methods

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Abstract

When wafers are stored in a plastic storage box to protect them from airborne contaminants, volatile organics from the polymeric construction material adsorb onto the wafer surface. Both UV/O_3 and ECR oxygen plasma techniques have been found to completely remove these organic contaminants adsorbed on the silicon surfaces up to the detection limit. After cleaning, Si wafers were characterized using attenuated total reflection-Fourier transform infrared spectroscopy (ATR-FTIR) and atomic force microscopy (AFM). Organic contaminants were eliminated more efficiently by the ECR oxygen plasma technique than the UV/O_3 technique when they were employed under optimum process conditions.
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Keywords: Dry cleaning; Organic contaminants; Electron cyclotron resonance (ECR) plasma; UV/O_3; Attenuated total reflection-Fourier transform infrared spectroscopy (ATR-FTIR); Atomic force microscopy (AFM)

1. Introduction

As the linewidth of microelectronic devices shrinks towards 0.1 μm, contaminants that have not traditionally been a processing problem become important issues. Sources of volatile organic contaminants on semiconductor wafers surfaces include the construction materials, compounds from fabrication processes and outgassing from the local environment. The effects [1] include changes in oxide quality, growth rate and adhesion. When silicon wafers are exposed to the atmosphere, even in a regulated clean room, gaseous organic molecules in the air are easily adsorbed by the wafer surfaces in a short time. To avoid the adsorption of these airborne organics, silicon wafers are, in general, stored in plastic boxes, which are usually made of polypropylene or polycarbonate materials. However, the problem of organic contamination still exists due to outgassing of contaminants from the plastic materials in the storage boxes [2].

Among the methods adopted in wafer cleaning technology, wet chemical methods based on RCA cleaning are widely used [3]. However, as devices are increasingly scaled down, the problems related to wet chemical cleaning become more serious. Among the dry cleaning methods, the most effective techniques for removing organic contaminants are UV cleaning and plasma-enhanced cleaning. It is worth noting [4] that UV/O_3 cleaning causes less radiation damage and is more controllable. In addition, plasma processes have been used for removing organic contamination, including remote oxygen plasma exposure. Although remote plasma oxidation works well, it suffers the disadvantage of subjecting the surface to energetic ion bombardment. Recently, magnetically confined microwave or electron cyclotron resonance (ECR) [5] plasmas have been explored for cleaning applications.

In this paper, comparison of the removal efficiency for organic contaminants on silicon wafers stored in plastic boxes between UV/O_3 and ECR oxygen plasma cleaning methods is reported.

2. Experimental

The silicon wafers used in this study were p-type, B-doped, Si(100), having dimensions of $50\times10\times3$ mm^3 (/45°). Prior to the intentional contamination, conven-

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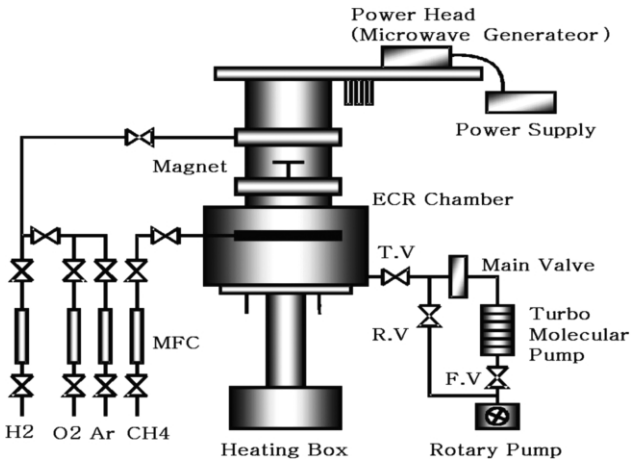


Fig. 1. Schematic diagram of ECR plasma system.

tional cleaning methods were used. Then 20 ml of pure acetone was placed in a wafer storage box and the wafers were intentionally contaminated with contaminated acetone vapor in a clean hood for 3 h. The contaminated wafers were separately transferred into UV/O_3 or ECR plasma chambers. The UV cleaning box contained low-pressure mercury discharge lamps at a wavelength of 253.7 nm. The samples were kept at a distance of 1 cm from the lamp. In ECR plasma using O_2, the samples were positioned 20 cm below the center of the electromagnet. The base pressure of the ECR plasma chamber was 5×10^{-6} Torr, while the process pressure was approximately 5×10^{-3} Torr with a flow rate of 10 sccm of oxygen gas. The microwave power was 300 W at a frequency of 2.45 GHz and the plasma chamber (Fig. 1) had the form of a cylinder, made of stainless steel, with a diameter of 160 mm and a height of approximately 150 mm. The Si wafers were characterized by ATR-FTIR (Nicolet 520 FTIR spectrometer). The incident angle was 45°, which resulted in 15 internal reflections (Fig. 2). Finally, the samples were characterized by AFM (Topomatrix Corporation: Accurex II).

3. Results and discussion

It is quite evident from Fig. 3 that the amount of organic contaminants decreases with increasing UV/O_3 exposure, and the time to reach the detection limit for

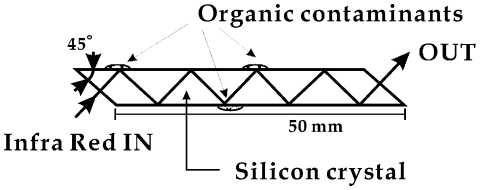


Fig. 2. Schematic representation of the ATR-FTIR method.

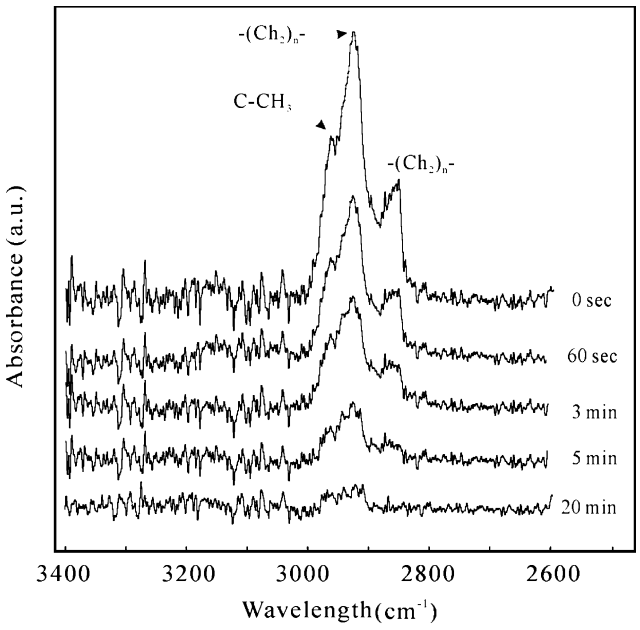


Fig. 3. ATR-FTIR spectra of the Si wafer after different exposure times during UV/O_3 cleaning.

ATR-FTIR is approximately 10 min. Fig. 3 shows ATR-FTIR spectra for the silicon wafers cleaned by UV/O_3 with different exposure times. The absorption spectra resulting from the organic contaminants are marked by several peaks, with C–CH_3 at 2960 cm^{-1}, –CH_2– at 2930 cm^{-1}, –CH at 2890 cm^{-1} and C–H stretching vibrations [–CH_3 and –$(CH_2)_n$–] at 2870 cm^{-1} [6]. The intensity of the peak at 2930 cm^{-1} is a maximum, implying the presence of –CH_2– stretching vibrations as the predominant group.

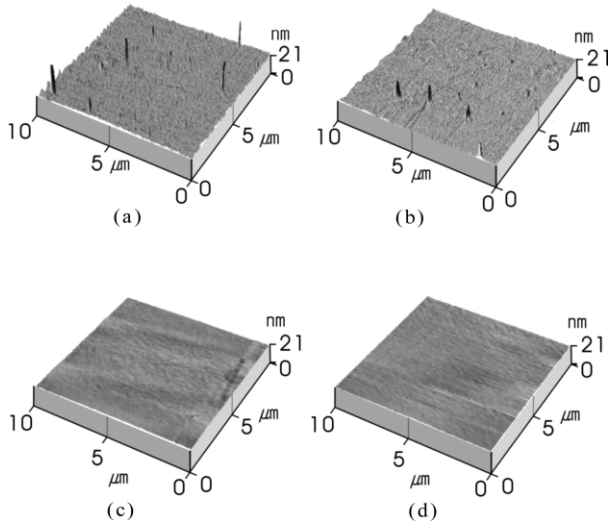


Fig. 4. Three-dimensional AFM images of Si wafer surface: (a) before exposure; and exposed to UV/O_3 for (b) 1; (c) 3; and (d) 5 min.

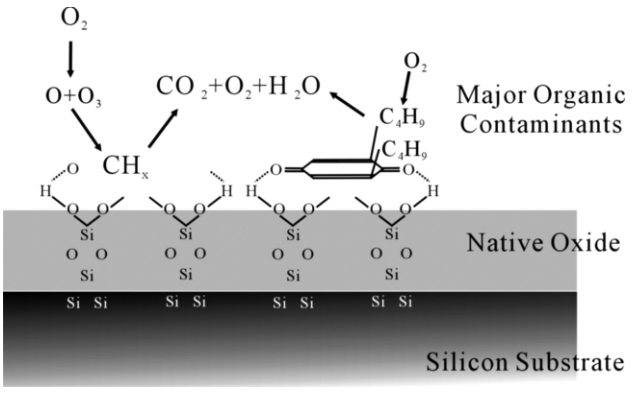


Fig. 5. Model for adsorption of organic contaminants on silicon wafers and the hydrocarbon (Si oxide) removal mechanism by the UV/O_3 cleaning process.

Fig. 4 shows three-dimensional AFM images of the Si surface as a function of UV/O_3 exposure time. The RMS roughness before UV/O_3 treatment was 0.97 nm; after UV/O_3 treatment for 1, 3 and 5 min at room temperature, this value decreased to 0.69, 0.49 and 0.36 nm, respectively. It may be noted here that the reduction rate for RMS roughness is steeper during the time interval of 0–1 min in comparison with later time intervals. Thus, the removal efficiency during the first 3 min of UV/O_3 treatment was much better than that after 3 min. Considering the fact that the presence of organic contaminants increases the surface roughness, we can explain this reduction in RMS surface roughness with increasing UV/O_3 exposure time by the removal of organic contaminants from the silicon wafer surface, as evident from Fig. 3.

Organic contaminants [7] desorbed from polypropylene storage boxes are mainly 2,6-di-*t*-butyl-2,5-cyclohexadiene-1,4-dione, dibutyl phthalate (DBP) and 2,6-di-*t*-butyl-4-methyl-phenol (BHT). In general, a silicon wafer surface shows strong polarity effects and therefore is suitable for attracting organic additives containing polar groups. As a result, organic contaminants that have relatively low vapor pressures, small molecular weights and polar groups, such as $>C=O$ and $-OH$, can easily be adsorbed by the native oxide of the silicon surface (Fig. 5).

It is generally accepted that hydrocarbon removal by UV/O_3 cleaning is primarily accomplished by photosensitized oxidation processes. Due to UV excitation, excited contaminant molecules and free radicals produced by the dissociation react with atomic oxygen to form simpler volatile molecules, such as CO_2 and H_2O [8]. The contaminants are removed through the lift-off process when the underlying oxide layers are etched off during the UV/O_3 treatment. Although UV/O_3 cleaning produces less radiation damage, as evident from the small RMS roughness values, and is more controllable, UV/O_3 treatment for a long time could cause deleterious effects for subsequent steps in device fabrication.

Fig. 6 shows ATR-FTIR spectra for the silicon wafers as a function of the ECR oxygen plasma exposure time. The presence of peaks similar to those in Fig. 3 can be associated with CH stretching vibrations, as discussed previously. While the UV/O_3 treatment took approximately 10 min to reach the ATR-FTIR detection limit, the ECR oxygen plasma treatment took approximately 40 s. Therefore, in contrast to the UV/O_3 cleaning, the ECR oxygen plasma treatment is successful in decreasing the process time for removing organic contaminants. The removal efficiency increases during the first 10 s of exposure, but decreases after 10 s.

Fig. 7 shows atomic force micrographs of Si wafers at various stages of the ECR oxygen plasma treatment. It is evident that smoothing of the silicon surface occurs after 10 s of exposure to the ECR oxygen plasma (Fig. 7b). However, the surface is later marked by the presence of pores on the silicon surface (Fig. 7c,d) after exposure of 50 s, and with increasing time the size of the pores becomes larger. The RMS roughness before the ECR oxygen plasma treatment was 2.32 nm. This relatively high value may be due to the difference in the level of organic contamination during preparation. The RMS roughness values after the ECR oxygen plasma treatment for 10, 20 and 30 s and 1 min at room temperature are 0.53, 1.33, 1.56 and 3.21 nm, respectively. This means that, as the plasma exposure time increases, the RMS roughness decreases during the first 10 s of exposure, but increases rapidly thereafter.

Hence, from the AFM analysis results, the optimum plasma exposure time seems to be approximately 10 s,

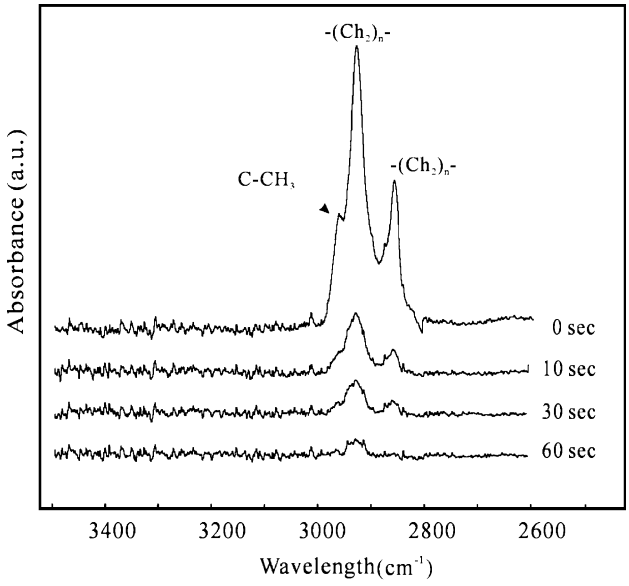


Fig. 6. ATR-FTIR spectra of Si wafers after different exposure times during ECR oxygen plasma cleaning.

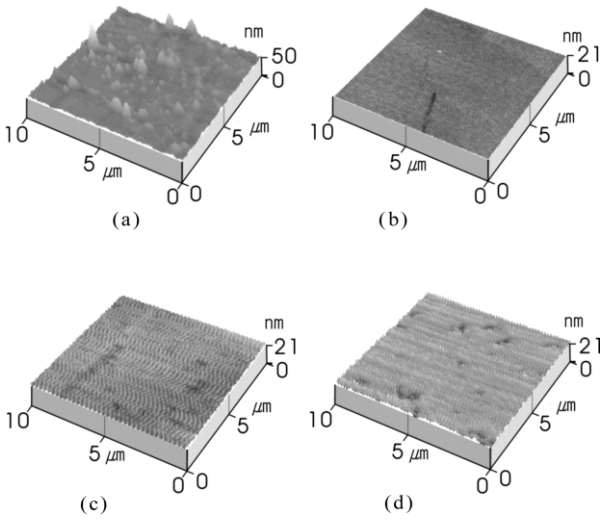


Fig. 7. Three-dimensional AFM images of Si wafer: (a) before exposure; and exposed to ECR oxygen plasma for (b) 10; (c) 50; and (d) 60 s.

since the surface roughness is found to increase after 10 s; however, organic contaminant removal is incomplete at 10 s, as is evident from the ATR-FTIR results (Fig. 6). Combining these two facts, an optimum exposure time of approximately 30–40 s is suggested.

The decrease in surface roughness with ECR oxygen plasma exposure time up to 10 s may be due to the removal of organic contaminants from the silicon surface. On the contrary, the increase in surface roughness with plasma exposure time after 10 s may be attributed to the result of oxygen ion bombardment of the silicon surface. This happens because excited oxygen atoms generated in O_2 plasma have greater molecular weight and higher energy than hydrogen atoms in H_2 plasma. In addition, the surface roughness is attributed to non-uniform oxidation of the silicon wafer surface during oxygen plasma exposure. When oxygen ions or radicals oxidize contaminants on the surface, the surfaces of the silicon wafers are also oxidized. After organic contaminants are removed by ECR plasma, the roughness of the silicon wafer surface is increased, causing substantial damage to the silicon surface.

There are two different views on the removal mechanism for organic contaminants using ECR oxygen plasma. One is that oxygen atoms are ionized in oxygen plasma. These oxygen ions bombard and remove organic contaminants on the silicon surface (sputtered off). The other is that some excited oxygen atoms in oxygen plasma adhere to the organic compounds, oxidize them, and finally decompose them into H_2O, O_2, H_2, CO and CO_2, all of which have high vapor pressures (lift-off).

On the basis of the above experimental facts, we propose the following mechanism for the removal of organic contaminants. Organic additives such as BHT that have polar groups, such as –OH and =O, are adsorbed on the silicon substrate via the polar –OH group of the surface native oxide, as shown in Fig. 5. These additives are adsorbed by the native oxide surface due to strong H-bonding. Under normal conditions, however, several layers of water molecules may be adsorbed by the native oxide due to H-bonding. In this case, the organic additives are adsorbed onto the H–O–H polar group of water molecules due to H-bonding, the same strong bonding as for the case without water molecules shown in Fig. 5. However, oxygen ion bombardment is strong enough to remove these organic contaminants from silicon wafers and the organic contaminants are chemisorbed by the silicon substrate.

To remove these organic contaminants ($C_xH_yO_z$), C–Si, H–Si and O–Si bonds must be broken. In other words the threshold ion energy is the same as the largest value of the bond strength for C–Si, H–Si and O–Si. In order to break these bonds the ion energy must be sufficiently greater than the bond strength. The values of these bond strengths are given in Table 1 [9].

The ion energy is higher near the exhaust and lower near the microwave window in an ECR plasma chamber. If we calculate the ion energy at 0.75×10^{-3} Torr, the value is in the range of 1446–3470.4 KJ/mol from the ion energy distribution [10]. This energy is sufficiently large to break the C–Si, H–Si and O–Si bonds (Table 1). Therefore, the energetic ions are capable of removing these organic contaminants from the silicon surface at the operating pressure of 5×10^{-3} Torr.

It is expected to cause the damage (pores) on the silicon wafer surface after adequate plasma exposure time. The pores seem to be either located at or adjacent to the locations of the organic contaminants. Electronic excitation due to ion bombardment leads to the extraction of oxygen atoms (O). After this extraction, some dangling bonds remain on the silicon surface. With subsequent ion bombardment the extracted O reacts with the dangling bonds and recreates the Si–O network. However, the network is deformed and strained in a metastable state. Si–O groups in metastable areas are easily removed, in contrast to Si–O in other areas. This leads to the formation of pores on the silicon surface. Therefore, despite the very effective role of ECR oxygen plasma cleaning in the quick removal of organic contam-

Table 1
The bond energy for C–Si, H–Si and O–Si

Bond	Bond energy	
	(kcal/mol)	(KJ/mol)
C–Si	104±5	435±21
O–Si	190.9±2	798.7±8.4
H–Si	71.34±0.11	298.49±0.46

inants from the silicon surface, the surface roughness increases with the plasma exposure time.

4. Conclusions

In this study, the removal efficiency for organic contaminants on (100) silicon wafers stored in plastic boxes was compared between UV/O_3 and ECR oxygen plasma cleaning methods. For the former, the UV/O_3 exposure time to reach the detection limit for ATR-FTIR was approximately 10 min and the surface roughness decreased with exposure time. The contaminants are likely to be removed through the lift-off mechanism when the underlying oxide layers are etched off during UV/O_3 cleaning. In ECR oxygen plasma cleaning, the plasma exposure time to reach both the detection limit and the lowest RMS roughness was 40 and 10 s, respectively. The optimum plasma exposure time is 30–40 s considering the effects of both cleaning and surface roughening. It is proposed that the cleaning mechanism is a combination of sputter-off and lift-off mechanisms following chemical reactions between the chemically reactive species and organic contaminants. The ECR oxygen plasma cleaning technique seems to be more effective than the UV/O_3 cleaning for the removal of organic contaminants, but it can cause substantial damage to the silicon surface.

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